

SDM234-Homework3

1. Are the following functions harmonic? If your answer is yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

(1) $u = x^2y + xy^2$;

(2) $u = x^3 - 3xy^2 + y$;

(3) $u = e^{3x} \cos 4y$;

(4) $u = \operatorname{Re}(\cos z)$;

(5) $v = \frac{x-1}{(x-1)^2 + y^2}, \quad f(2) = i$.

2. Show that if u is harmonic and v is a harmonic conjugate of u , then u is a harmonic conjugate of $-v$.

3. Find the value and principal value of:

(1) $\ln((1+i)^2)$;

(2) $\ln\left(\frac{1-i}{1+i}\right)$;

(3) $(1+i)^i$;

(4) $(-e)^{1+i}$.

4. Find the value of:

(1) $e^{\ln 2 - \frac{\pi}{2}i}$;

(2) $\cosh\left(\ln 2 + \frac{\pi}{2}i\right)$;

(3) $\cos\left(\frac{\pi}{2} + i \ln 2\right)$.

5. Solve the following equations:

(1) $\sin z = 2$;

(2) $\cosh z = i$.

6. Prove the following product to sum formula and sum to product formula for trigonometric and hyperbolic functions:

(1) $\sin z_1 \cos z_2 = \frac{1}{2}[\sin(z_1 + z_2) + \sin(z_1 - z_2)]$;

(2) $\sin z_1 + \sin z_2 = 2 \sin\left(\frac{z_1 + z_2}{2}\right) \cos\left(\frac{z_1 - z_2}{2}\right)$;

(3) $\sinh x \cdot \sinh y = \frac{1}{2}[\cosh(x + y) - \cosh(x - y)]$;

(4) $\sinh x + \sinh y = 2 \sinh\left(\frac{x + y}{2}\right) \cosh\left(\frac{x - y}{2}\right)$.